Frontier



Fusion Carbon Sequestration

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Fusion Carbon Sequestration, LLC

Organization location (we welcome applicants from anywhere in the world)

Somerville, MA, USA

Name(s) of primary point(s) of contact for this application

Andy Haverly

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

For-profit organization affiliated with Massachusetts Institute of Technology.

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Subsurface blasts to improve our understanding of using explosives for rock comminution for large-scale enhanced rock weathering to sequester carbon.

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Enhanced rock weathering (ERW) is one of the most promising methods for carbon dioxide removal at gigaton scales. One of the most difficult challenges for ERW is the difficulty of rock comminution. The current techniques for rock comminution involve crushers, SAG mills, and ball mills. Each of these machines is expensive and extremely inefficient. To put it into perspective, it requires less energy to melt the rocks than it does to use ball mills to grind the rocks into small particles. It is also orders of magnitude more energy intensive than the Gibbs free energy of the rock.

We are focusing on skipping these inefficient steps. With in-situ comminution we can perform larger blasts with higher energy efficiency, with lower capital and operational costs, in a more scalable manner. Each of the problems that are faced by current rock grinding processes are addressed to some extent with subsurface comminution blasts.

It should be emphasized how scalable explosives truly are. The cost of conventional mining explosives, like ANFO, scales linearly with the amount of chemical energy input into them. The most energetic things that humans have ever created were fusion explosives. There is plenty of design and research around the next generation of explosives that use just fusion energy to release more energy than is put in. Explosives can be scaled infinitely large with current technology, for very little increase in cost. Whereas all other forms of comminution require approximately linear increases in cost. This is especially relevant at large scales where this near-zero scaling cost of explosives outcompetes the linear scaling cost of every other form of comminution.

Fusion explosives encapsulate a large class of explosives. The most notable ones are infamous: thermonuclear explosives. However, there are other explosives that use fusion, including inertial confinement fusion (ICF) and other similar approaches that all create high pressure and high temperature environments that enable fusion. While we want to acknowledge that thermonuclear explosives are technically explosives that can be used here we will be generalizing fusion explosives to encapsulate the safer alternatives. Fusion explosives are primarily composed of two parts: the primary and the fusion explosive. The primary is the force that enables the fusion fuel to undergo fusion. Once the fuel is able to undergo fusion, the cost per yield drops dramatically. Castle SHRIMP produced 10,000 tons of TNT equivalent per kilogram of lithium deuteride (LiD). LiD costs around $100/kg. This amounts to $0.00001/kg TNT equivalent whereas conventional explosives cost ~$1/kg. Therefore, while conventional explosives are cheaper for small scale blasts, fusion explosives are unmatched for their scalability.

The step-change advancement that this project aims to achieve is to answer whether or not explosives can comminute rocks in an efficient and scalable manner. If this capability is possible, then ERW will truly be unlocked. This will change the bottleneck for ERW from rock comminution to rock spreading. This advancement will completely remove the largest bottleneck for ERW. Once this largest bottleneck is removed, the rest of the bottlenecks are easier to solve.

Additionally, there is a significant amount of fines produced from blasting that are too large for efficient carbon sequestration. These fines are angular and can be used in concrete as fine aggregate. Due to the massive scale of the amount of fines that will need to be produced to sequester the gigatons of carbon dioxide necessary to offset global emissions, we will also produce

gigatons of fines that can be sold for use in concrete. Depending on the type of explosives used, their cost, and their scale, we can potentially produce enough value from these fines to offset the cost of carbon dioxide sequestration and actually make climate change mitigation profitable, even without carbon credits in the future.

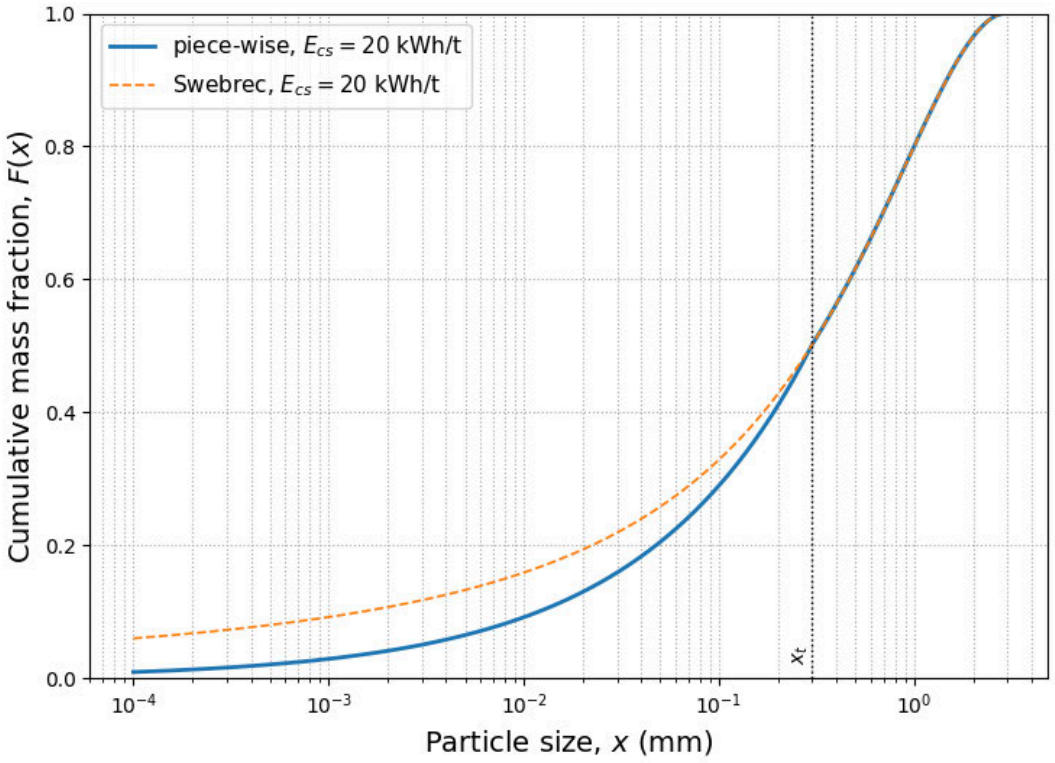
Our modeling shows that the entire process of using subsurface blasts using fusion explosives, without considering the cost of spreading the rocks, costs $0.01 per ton of CO_2 sequestered. If we consider the fact that the byproducts can be sold for a profit then this process becomes profitable. The energy costs for this comminution are almost entirely for the production of the primary that triggers the fusion process, given that we have designed the rest of the process to minimize costs. Because fusion explosives are profitable for comminution, it is hard to compare their cost to other forms of comminution. When we consider conventional explosives, the costs become more comparable to existing forms of comminution. We estimate that it will cost ~$32/ton of CO_2 sequestered, just for rock comminution. This accounts for the carbon needed to be offset from the explosives. This price is more comparable to existing rock pulverization with ball mills ($150-300/ton of CO_2 sequestered). It should be noted that there is plenty of variability in this cost because there is currently enough mining tailings that can be used for ERW until the scale approaches the necessary scale to offset our global emissions.

Therefore, our core research goal is to quantify the efficiency of highly confined blasting for rock comminution. We would like to perform an underground blast then sample the surrounding rocks using a core drill. With this experiment we will be able to quantify the particle size distribution (PSD) vs. specific comminution energy (Ecs) and Ecs vs. distance relationships that are necessary to state whether or not blasting will scale efficiently to meet the gigaton demand for ERW, with any type of explosive.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

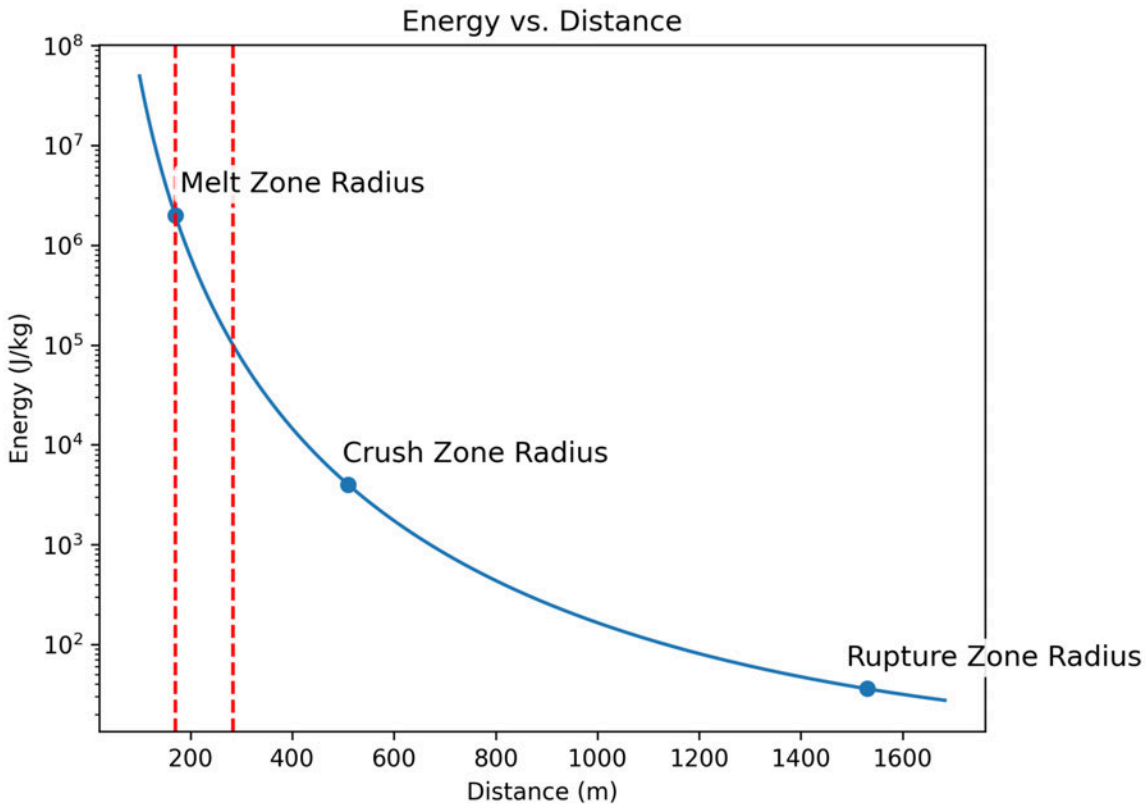
Large-scale ERW requires large amounts of rocks that are in a fine powder form. As such, we aim to refine two models from this project: particle size distribution (PSD) as a function of specific comminution energy (Ecs) and Ecs as a function of distance from the blast. Both of these questions can be answered from this project.

First, we will discuss the PSD vs. Ecs relationship. The industry standard model for the PSD for fines generation is the Swebrec model. This model has been used for many years and is accurate for typical blasting regimes. However, this model is not tested or measured for high Ecs regimes. Below is a model showing both the Swebrec model as well as our own model that combines the Swebrec model with the branch-and-merge model for high-energy fragmentation.

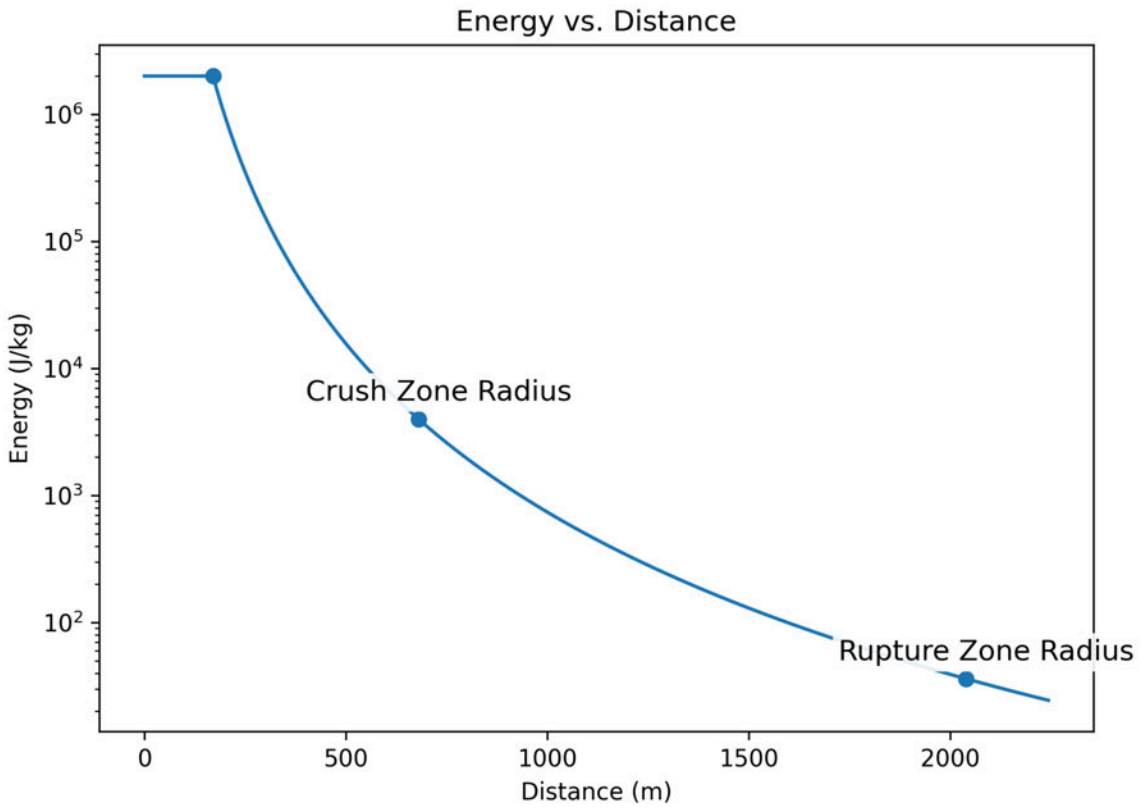


This combined model shows that the Swebrec model likely overestimates the amount of fines produced (more details are available in the pre-print). It is important to note that for these particles to sequester carbon in reasonable time-scales they must be smaller than approximately 10 microns. To understand how many carbon-sequestering fines we are able to produce from a blast, we must have a better experimental understanding of this PSD vs. Ecs model.

The next relationship that we must understand is the relationship between Ecs and distance from the blast. Because the PSD is a direct result of the Ecs, understanding the Ecs as a function of distance from the blast will allow us to use the PSD vs. Ecs model to quantify the amount of fines produced from a blast. A model for the Ecs vs. distance relationship is shown below.

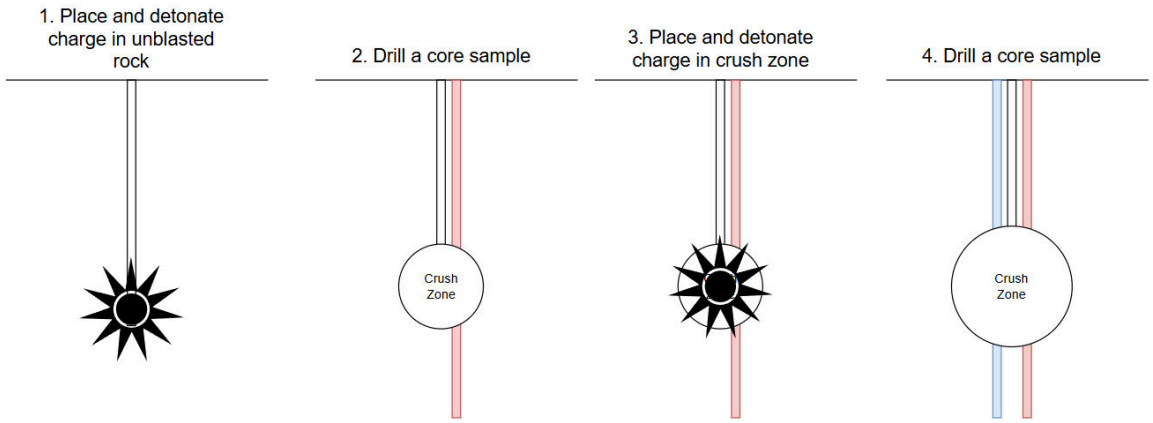


This Ecs vs. distance model is shown for a very large blast. In this model we can see that at very close distances from the blast the energy is high enough to melt the rocks. Then the energy falls off super-exponentially, indicating that almost all of the fines are generated very close to the blast. This is logically consistent. To address the issue of rocks melting, we can incorporate a material that turns this extra heat into mechanical shock: water. Water allows the shock to propagate while keeping the temperature low enough that the surrounding rocks do not melt. This is shown below.



When we combine the PSD vs. Ecs and Ecs vs. distance models we can create a single function that quantifies the amount of carbon-sequestering rock produced from the energy of the blast. This single function allows us to quantify how many explosions are needed to produce the gigatons of comminuted rocks for gigatons of carbon sequestration.

All of these models are created from publicly available data, but the data itself does not provide the answers to the question: can explosives be used for large-scale rock comminution in a scalable and economic manner. Therefore, we have designed some blasts to answer this question. This is shown below.



This figure shows two blasts with two core samples. This blast design uses standard blasting equipment and processes, so any mining company can implement this blast quickly. The first blast and core sample will allow us to extract both the PSD vs. Ecs and the Ecs vs. distance relationships for blasting in virgin rock. We can extract this information by measuring the PSD at various points in the core sample. From the PSD we can use the Swebrec model to quantify the Ecs, as well as have an experimental understanding of the finer tail of the PSD. Once we know the Ecs of a point in a core sample, we can measure the distance of the core sample from the blast and have the Ecs vs. distance relationship. Therefore, this single blast and core sample process will provide experimental values that will allow us to quantify the amount of fines created from a blast of any size.

There are many parameters that can be changed to improve the efficiency of blasting for a particular use case. Some of these parameters include larger charges, multiple charges, repeated blasts, different explosives with different shear velocities, etc. These parameters are well studied for smaller Ecs, but not well studied for this high energy regime. As an example of why understanding this is useful, let's compare a single charge with two charges in close proximity. In the single charge scenario, the energy is radiated outward. However, in the two charge scenario, the shock waves constructively and destructively interfere. This means that the effective instantaneous Ecs that the rock between the charges sees is much larger than if the charges were detonated alone. This enhanced Ecs may provide a much higher fraction of fines.

In addition to understanding the blasting in virgin rock, we also want to quantify the fines generated from multiple blasts. To do this, we can reuse the blast hole, detonate more explosives, then take another core sample. We expect this blast to further shift the PSD to the finer regime. Reusing the holes like this is economical both for this test and as we scale this form of carbon sequestration because we can reuse the hole, which is one of the most expensive parts of this implementation at larger scales.

It should be noted that, while optimizing the particle size distribution from these blasts is important, optimization is likely overemphasized. Fusion explosives cost $0.00001 per kg TNT equivalent. The added cost of improving the efficiency is likely not worth it. It makes more sense to just increase the yield of the explosive. The fuel cost difference between a megaton explosive and a gigaton explosive is $10 million. That is a negligible cost difference at these scales.

We plan to do this blast at the Colorado School of Mines Edgar Experimental Mine. We have spoken with the coordinators at this mine. They have the necessary equipment: drill, blasting equipment, etc.

required for standard blasts like this. We expect to do a blast in the 20-80 lbs of TNT equivalent range. Once the core samples are taken, we will take them to MIT. At MIT, we have the equipment necessary to characterize these samples. We have automatic optical microscopes and automatic scanning electron microscopes that have been used to quantify the PSD for previous blasts and can be used for future blasts. This automatic characterization system allows for high accuracy and high throughput characterization.

Because we are doing such small blasts, we expect less than a ton of fines to be produced and most of the fines to remain underground and not sequester carbon dioxide. We do not expect any notable amount of carbon sequestration from this project. However, we do expect that if this project is successful that we can massively scale up the amount of fines that can be produced for ERW from blasting. The vast majority of the cost of this project is not on explosives; ANFO costs ~$1/lb. This means that future studies with larger conventional blasts scale well.

Fusion explosives are under a lot of study right now, mostly in terms of inertial confinement fusion. The research behind these explosives is important not only for energy production, but also for the mechanical energy they can produce. The sun produces all of its energy from fusion and the research behind fusion explosives is rapidly advancing in recent years. We would like to also study how fusion explosives can be used for rock comminution for carbon sequestration. We expect that there is little carbon footprint for these explosives compared to conventional explosives. However, there will be tradeoffs for fusion explosives, mainly in terms of X-rays, gamma rays, and fast neutrons, as well as neutron activation and fission products depending on the specifications of the fusion explosives. We believe that the X-rays, gamma rays, and fast neutrons can be turned into useful mechanical shock by superheating water. The neutron activation and fission products are longer-lived products that we would like to analyze properly. We believe that all of the advances in fusion explosives will provide an excellent long-term approach to cheaper and more accessible carbon sequestration.

This method of advancing ERW is unique because there is no one else studying the utility of blasting for carbon sequestration. All other ERW projects focus on a different part of the ERW process or use a less efficient and less scalable method of comminuting rocks. This project is unique and possesses the potential to enable gigatons of carbon sequestration in a scalable manner that every other form of carbon sequestration will struggle to match.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

The biggest risk of this project is inefficiency. There is no technical hurdle or unknown process that we must overcome for this experiment. We have designed our project around standard mining practices, using standard tools, equipment, and explosives. The only possible risk is that the explosives do not produce as many fines as predicted.

We will mitigate this risk by varying the parameters of the blasts to optimize the fines production. This includes using different explosives (e.g. fusion explosives), multiple simultaneous blasts with constructively interfering shock waves, using repeat blasts, varying the size of the charges, varying the spacing between the charges, etc. There is no limit to the number of parameters that can be

changed to produce better results from these blasts.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	Multiple blasts and core samples, measured data on the PSD vs. Ecs and Ecs vs. distance for samples along the cores, models for optimizing and estimating the mass of comminuted particles, and scaling estimates for larger explosions using different types of explosives.
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	Enhanced rock weathering requires large amounts of pulverized rocks and grinding the rocks is far too expensive. If this project is successful, we can simplify, reduce the cost, accelerate the processing, and enable much larger scale rock pulverization for enhanced rock weathering.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	The core deliverables are performing the underground blast (1 month), measuring the PSD and Ecs from the core samples (2 months), optimizing the blasts with multiple charges (7 days), optimizing repeat blasts (9 days), optimizing larger blasts (1 month), and optimizing fusion blasts (2 months).
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$250k: $200k for blasting, core sampling, and analysis, $50k for blast optimization and extrapolation.